shooting out of galactic nuclei, about the gigantic, magnetized intergalactic lobes that the jets feed, and about the magnetized beams of pulsars. The X-rays, with their thousand-fold shorter wavelengths than light, were produced mostly by high-speed electrons in ultra-hot gas accreting onto black holes and neutron stars, and they thus taught us directly about the accreting gas and indirectly about the holes and neutron stars.

The differences between light, on the one hand, and radio waves and X-rays, on the other, are pale compared to the differences between the electromagnetic waves (light, radio, infrared, ultraviolet, X-ray, and gamma ray) of modern astronomy and gravitational waves. Correspondingly, gravitational waves might revolutionize our understanding of the Universe even more than did radio waves and X-rays. Among the differences between electromagnetic waves and gravitational waves, and their consequences, are these[6]:

- The gravitational waves should be emitted most strongly by large-scale, coherent vibrations of spacetime curvature (for example, the collision and coalescence of two black holes) and by large-scale, coherent motions of huge amounts of matter (for example, the implosion of the core of a star that triggers a supernova, or the inspiral and merger of two neutron stars that are orbiting each other). Therefore, gravitational waves should show us the motions of huge curvatures and huge masses. By contrast, cosmic electromagnetic waves are usually emitted individually and separately by enormous numbers of individual and separate atoms or electrons; and these individual electromagnetic waves, each oscillating in a slightly different manner, then superimpose on each other to produce the total wave that an astronomer measures. As a result, from electromagnetic waves we learn primarily about the temperature, density, and magnetic fields experienced by the emitting atoms and electrons.
- Gravitational waves are emitted most strongly in regions of space where gravity is so intense that Newton’s description fails and must be replaced by Einstein’s, and where huge amounts of matter or spacetime curvature move or vibrate or swirl at near the speed of light. Examples are the big bang origin of the Universe, the collisions of black holes, and the pulsations of newborn neutron stars